

Wave

MP

Teach You Physics Hand In Hand

March 2026

*Please make sure that you know this, as it is **frequently** asked in DSE.*

1 The Wave Equation

By applying $F = ma$ to an infinitesimal segment of a wave:

$$F\left[\left(\frac{\partial y}{\partial x}\right)_{x+\delta x} - \left(\frac{\partial y}{\partial x}\right)_x\right] = \mu\delta x \frac{\partial^2 y}{\partial t^2}$$

We can get the Wave Equation, which is a **partial differential equation**:

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2} = \frac{\mu}{F} \frac{\partial^2 y}{\partial t^2}$$

2 The Wave Function

'Wave Function (ψ)' means '**a general solution to the Wave Equation**'. It has a general form of:

$$\psi(x, t) = f(x - vt) + g(x + vt)$$

You always encounter sinusoidal wave in DSE, which specifies the general form of the Wave Function into a form of:

$$\psi(x, t) = A \cos(kx - \omega t)$$

where **wave constant** $k = \frac{2\pi}{\lambda}$

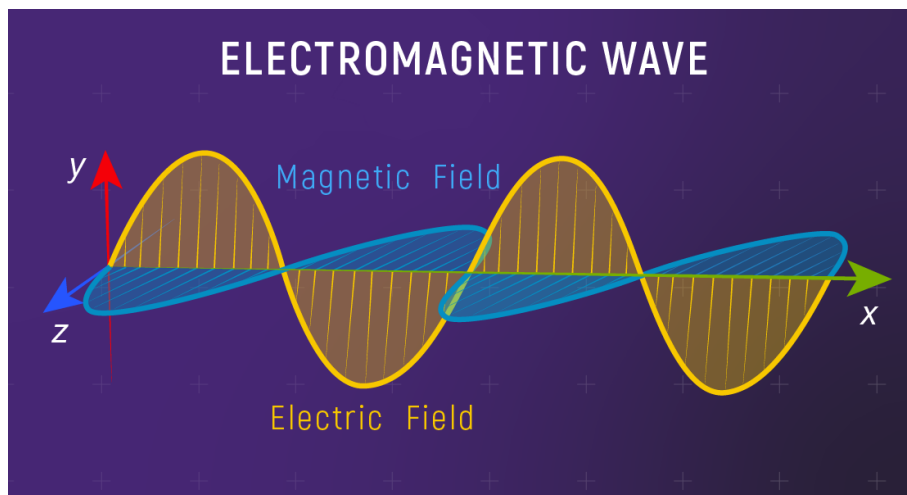
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3 Problem

Problem:

Given that light is composed of a wave-form electric field and a wave-form magnetic field. Their maximum amplitudes E_0 and B_0 are at right angle to each other. The propagation direction of light can be found by the Right-hand-grip Rule: curling your 4 fingers from E_0 to B_0 , and your thumb gives the direction of propagation.

The incident light has an electric field in the form of $\vec{E}(x, t) = \hat{y}E_0 \sin(kx - \omega t)$. What is the wave form of the magnetic field in the normally reflected light?



Solution will be uploaded in the next post and MP love you all :)